



Fig. 3: Wiring diagram of extension cable

tronic fuse box for the car. At the heart of the circuit is a medium-power silicon epitaxial-base npn power transistor in monolithic Darlington configuration BD677 (T2).

Transistor T1 functions as a constant current generator, with its base held at a constant voltage by LED2. Consequently, the emitter resistor R3 supplies a constant emitter (and collector) current of nearly 12 mA. Since SCR1 feeds the base current into transistor T2, normally the output load gets powered. LED1 indicates the presence of a power supply.

Fig. 1 shows the circuit diagram of the electronic fuse and Fig. 2 shows the

pin configuration of the components. As shown in Fig. 1, the output current to the load flows through resistor R4. As the load current rises to a value of 1A, a voltage drop of about 0.8V occurs across resistor R4, which fires SCR1. As a result, the base of transistor T2

is pulled down to ground and transistor T2 is cut off to break the current flow through the output load. The circuit remains in this state until the input supply is interrupted. Switch S1 is pressed to reset the circuit. Before resetting the fuse, any short circuit condition should be taken care of, otherwise SCRT1 will fire again to switch off the load through transistor T2.

It is simple to calculate the current which activates the circuit. As already explained, the circuit is activated at a voltage of 0.8V across R4. Therefore, the electronic fuse comes into action at a current of 1A, with $R4=0.82\text{-ohm}$.

Similarly, the actuation current (fuse rating) is 500 mA with 1.6 ohm (or 250 mA with 3.2-ohm). Since tolerance in the value of resistance may change the load current, the value of resistor R4 can only be calculated by trial and error. Alternatively, a wirewound power potentiometer (10 ohm, 5W) can be used in place of R4.

The fuse circuit is assembled on a general-purpose PCB and placed in a suitable plastic enclosure as shown in Fig. 3. Connect a 12V car battery supply (available through the standard 12V DC car power socket) to the circuit through a cable as input. Use a two-pin connector on the front panel of the enclosure for supply to the gadgets.

EFY note. Car batteries are capable of supplying extremely high currents which can cause fire and other hazards. The circuit described here is only suitable for 12V car batteries. Never connect the negative terminal output lead of this fuse to the body of the vehicle. Values of all components indicated are critical for safe operation of the electronic fuse. ●

SOUND/VIBRATION-SENSITIVE TOY DOG ALARM

■ **PRADEEP G.**

Here is a simple circuit of a sound/vibration-sensitive toy dog alarm. In this

circuit, a condenser microphone is used to function as the ear of the dog, the speaker functions as the mouth and two LEDs function as the eyes. When the microphone

